Extended Vision Transformer (ExViT) for Land Use and Land Cover Classification: A Multimodal Deep Learning Framework

Jing Yao, *Member, IEEE*, Bing Zhang, *Fellow, IEEE*, Chenyu Li, *Student Member, IEEE*, Danfeng Hong, *Senior Member, IEEE*, and Jocelyn Chanussot, *Fellow, IEEE*

***Abstract*— The recent success of attention mechanism-driven deep models, like vision transformer (ViT) as one of the most representatives, has intrigued a wave of advanced research to explore their adaptation to broader domains. However, current transformer-based approaches in the remote sensing (RS) community pay more attention to single-modality data, which might lose expandability in making full use of the ever-growing multimodal Earth observation data. To this end, we propose a novel multimodal deep learning framework by extending conventional ViT with minimal modifications, aiming at the task of land use and land cover (LULC) classification. Unlike common stems that adopt either linear patch projection or deep regional embedder, our approach processes multimodal RS image patches with parallel branches of position-shared ViTs extended with separable convolution modules, which offers an economical solution to leverage both spatial and modality-specific channel information. Furthermore, to promote information exchange across heterogeneous modalities, their tokenized embeddings are then fused through a cross-modality attention (CMA) module by exploiting pixel-level spatial correlation in RS scenes. Both of these modifications significantly improve the discriminative ability of classification tokens in each modality and thus further performance increase can be finally attained by a full token-based decision-level fusion module. We conduct extensive experiments on two multimodal RS benchmark datasets, i.e., the Houston2013 dataset containing hyperspectral (HS) and light detection and ranging (LiDAR) data, and Berlin dataset with HS and synthetic aperture radar (SAR) data, to demonstrate that our extended vision transformer (ExViT) outperforms concurrent competitors based on transformer or convolutional neural network (CNN) backbones, in addition to several competitive machine-learning-based models. The source codes and investigated datasets of this work will be made publicly available at https://github.com/jingyao16/ExViT.**

***Index Terms*— Convolutional neural network (CNN), deep learning, hyperspectral, land use and land cover (LULC), light detection and ranging (LiDAR), multimodal image classification, synthetic aperture radar (SAR), vision transformer (ViT).**

I. Introduction

THE last decade has witnessed the rapid development achieved in aeronautics and astronautics, as well as remote sensing (RS) technology, providing geoscience analysts with a deluge of multisource and multimodal RS data [1]. The increasing availability of these data not only opens up new possibilities for a better understanding of intelligent Earth observation but also poses new methodological challenges in a wide variety of RS applications, such as ground material identification [2], [3], mineral exploration and mapping [4], scene understanding [5], [6], object detection [7], and change monitoring [8], [9]. In this article, we particularly focus on one of the most challenging tasks of land use and land cover (LULC) classification [10], which has been growing significantly vital in satisfying the ever-demanding need for precision agriculture and forestry, urban planning, environmental monitoring, and disaster response across industries and social governance.

Given one remotely observed image, the aim of LULC classification is to assign a unique label to each pixel according to its ground cover information with either man-made or natural causes. In practice, hyperspectral (HS) image has become one of the most concerning types of RS data when applied to LULC classification task, owing to its superior ability in capturing abundant spectral information that can precisely characterize the ground objects of interest [11]. For years, researchers have constantly focused on developing more effective feature extractors for HS images: for instance, Tuia et al. [12] defined an active set feature learner under a hierarchical setting, allowing for generating complex filters that can better explain the nonlinearities underlying HS data; Hsu [13] adopted the wavelet transformation and matching pursuit to extract robust HS image features for classification; and Ghamisi et al. [14] proposed a novel attribute profile-based method that is able to learn spatial invariant features and thus improve the classification performance. The diversified examples of this sort cannot be successfully put forward without resorting to advanced machine learning and image processing techniques.

With the aforementioned elaborately designed feature extractors in hand, the potential of HS data in identifying

Manuscript received 4 January 2023; revised 4 May 2023; accepted 28 May 2023. Date of publication 9 June 2023; date of current version 21 June 2023. This work was supported by the National Key Research and Development Program of China under Grant 2021YFB3900502. *(Corresponding author: Bing Zhang.)*

Jing Yao and Danfeng Hong are with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China (e-mail: yaojing@aircas.ac.cn; hongdf@aircas.ac.cn).

Bing Zhang is with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China, and also with the College of Resources and Environment, University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: zb@radi.ac.cn).

Chenyu Li is with the School of Mathematics, Southeast University, Nanjing 210096, China (e-mail: lichenyu@seu.edu.cn).

Jocelyn Chanussot is with CNRS, Grenoble INP, GIPSA-Laboratory, Université Grenoble Alpes, 38000 Grenoble, France, and also with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China (e-mail: jocelyn@hi.is).

Digital Object Identifier 10.1109/TGRS.2023.3284671

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:43:18 UTC from IEEE Xplore. Restrictions apply.

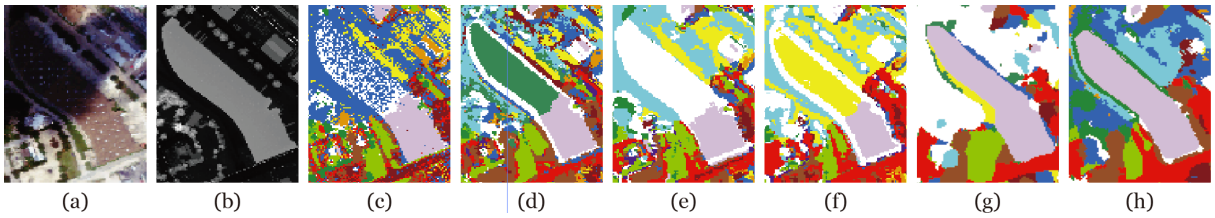


Fig. 1. Example that illustrates the superior classification performance achieved by our method in mapping LULC categories with multimodal RS image inputs. (a) and (b) False-color visualization of cloudy HS images and LiDAR-derived DSM, respectively. (c)–(g) Classification mapping results obtained by the representative concurrent multimodal classification methods, i.e., CoSpace [15], CCR-Net [16], and RS transformers [17], [18], however, in which the integrity of ground object cannot be well preserved due to their insufficient exploitation of complementary multimodal correlations under cloud or shadow disturbance. An observable improvement regarding this issue can be drawn from the result of our proposed ExViT in (h).

mixed pixels at the material level is largely boosted. However, relying on such a passive HS sensing mode only is still susceptible to the influence of volatile imaging conditions, such as changing illumination and cloud perturbation, and thus cannot guarantee an accurate and stable LULC mapping result. In this context, some latest methodologies have contrived a way to additionally take advantage of other modalities acquired by active instruments. Thanks to the complementary cues underlying heterogeneous modalities, like height information contained in light detection and ranging (LiDAR) data and geometric information contained in synthetic aperture radar (SAR) data, those pixels with not only indiscernible patterns from optical perspective but also different semantics but similar spatial and spectral patterns can get a chance to be distinguished, as the evidence in Fig. 1 shows.

Since deep learning has prevailed in the last few years, the conventional way of separate feature extraction and decision-making was revolutionized through broad attempts to utilize end-to-end neural networks with distinctive architectures and learning strategies. For example, Paoletti et al. [19] made an earlier attempt of constructing a novel spatial–spectral convolutional neural network (CNN) with efficient implementation for fast classification of HS images. Hang et al. [20] developed a patch-based reconstruction task-aided generative adversarial network method by mining rich unlabeled information in RS scenes. Hong et al. [21] addressed the problem of large-scale graph computation burden in leveraging non-grid relation regularization and proposed a mini version of graph convolutional network for efficient HS image classification. Inspired by the attention-like mechanism that mimics cognitive intelligence in neural science [22], there have been primitive attempt by introducing the attentional guidance into conventional CNNs [23] and more recent efforts on adjusting transformer architectures for RS tasks, owing to its salient benefit in modeling long-term dependences. Besides, the ideas of uncertainty-based active learning and physically constrained transfer learning were also considered, respectively, to alleviate the limited training sample issue that bothers "data hungry" deep classifiers [24], [25].

However, despite the remarkable improvements achieved by the aforementioned methods, there still exist critical issues that need to be pointed out on the way to a more accurate LULC mapping. To name a few, the practicality and transferability of advanced single-modality models remain restricted by their dedicated modular design, whereas the existing deep multimodal methods seldom model cross-modality dependences efficiently, thus hindering their further break of performance bottleneck. To this end, we propose in this article a general multimodal transformer framework for better information fusion in classifying multimodal RS data according to their LULC attributes. Unlike conventional linear projection and deep regional embedding-based tokenizor, we build our approach on a lightweight separable convolutional stem to process multimodal image patches by local and global perceptions along spatial and channel dimensions, respectively, as demonstrated in Fig. 2. A hybrid spatial vision transformer (ViT) backbone implemented with both self-attention and cross-modality attention (CMA) mechanisms is then designed to simultaneously draw intra and intermodality long-range correlations among tokenized features. To further encourage the modality interaction of heterogeneous data, the modality-specific classification tokens are fused before the multilayer perceptron (MLP) head, which can get rid of deciding the primary modality (branch) and aggregate discriminative information from multimodalities for the final decision. As expected, through this minimal but rational extension we made on ViTs, the classification results of the proposed method, as Fig. 1 shows, are substantially improved both in terms of quantitative accuracy and visual fidelity. More concretely, our key contributions in this article are summarized as follows.

1) Extended vision transformer (ExViT) is proposed as a novel multimodal DL framework for LULC classification of RS images. It can achieve desirable performance based on classic grid features while exhibiting an uncluttered and generalized architecture to follow.
2) We address the information exchange between multimodal RS by exploring the CMA that matches the encoding style of vanilla spatial ViT, in which the tokenized embeddings of heterogeneous modalities can be efficiently fused at levels from early to late to boost their discriminative ability.
3) To tackle the modality-incomplete issue in real RS scenarios, we establish a cross-modal multitask learning regime for our framework that is not only more interpretable than previous manners but also easy to implement.
4) Through extensive ablation studies, we empirically verify that replacing linear patch projection with plain separable convolution stem allows ViT backbone to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:43:18 UTC from IEEE Xplore. Restrictions apply.

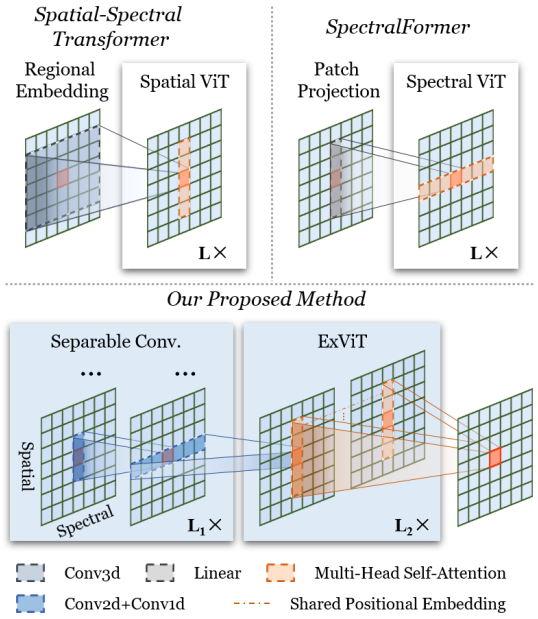


Fig. 2. Illustration that shows the difference in network connectivity between the ViT-based representatives and our proposed ExViT, which shows good modality-agnostic expandability with less requirement for the amount of training data.

extract more expressive spatial-channel representation (especially for HS images) without adding evident computational loads, and classification tokens are not necessary by substituting a simple yet effective all-tokens fusion prehead to further fuse the multimodal high-level semantics.

We present the whole organization of this article as follows. The research background and relevant existing work are introduced in Sections I and II. Then, Section III unfolds the proposed ExViT method, including a dual branch of separable convolution module extended ViT, a cross-attention module, and a token-fusion module. Experimental setting and LULC classification results are detailed in Section IV, with analysis based on the ablation study and comparison with representative conventional and deep methods on two benchmark multimodal RS datasets. We finally conclude this work with possible future directions in Section V.

II. Related Work

A. Multimodal LULC Classification: From Shallow to Deep

The ever-growing availability of multimodal RS data has intrigued a wide study on multimodal LULC classification. One typical type of conventional multimodal feature extraction method is built on manifold alignment regularized learning of linear projection into common subspace; Hong et al. [26] proposed a shared and specific feature learning model with modality-specific parameterization, which owns clear interpretability and yields great potential in LULC classification. Pournemat et al. [27] combined the idea of manifold local patching with joint spectral analysis of graph Laplacians of different modalities for semisupervised classification. The widely studied morphological profile and its variants, such as attribute profiles and extinction profiles [28], [29], were also introduced to jointly extract hierarchical features of multimodal input images. On the other hand, deep learning techniques have also found their place in handling multimodal RS data for their powerful ability in learning compact multimodal representation. A general multimodal network framework was established by Hong et al. [30] for the first time, which has explored extensive fusion strategies for heterogeneous RS data. The same investigators also devised a multimodal generative adversarial network-based method to circumvent the inherent problem of distribution discrepancy and adversarial disturbance under cross-sensor setting [31]. Inspired by the brain-like splitting autoencoder [32], Wu et al. [16] designed a multimodal CNN with cross-channel reconstruction for a more accurate joint classification of multimodal RS data. More advanced intelligent ideas such as collaborative and self-supervised learning have been recently introduced and lead to representatives, e.g., a plain coupled encoder–decoder backbone with embeds various kinds of domain knowledge by Li et al. [33], Ma et al. [34] elaborately presented dominant and interactive feature learning modules from a different modeling perspective, hoping to alleviate the semantic deviation of multimodality features, and Xue et al. [35] proposed a masked multimodal self-supervised training scheme to learn key feature representation.

B. Transformer-Based LULC Classification: From Single-Modality to Multimodal

The last several years have undoubtedly become the milestone when transformer-based models pervaded a broad range of visual perception and language recognition domains. However, it was not long past since they first cast a new light on the RS application of LULC classification. Among the newly proposed seminal works along this line, SpectralFormer [17], as the name suggested, is the first work to uncover HS images from a sequential perspective within the spectral domain. Qing et al. [36] also prepended a spectral attention module to extract more information from spectral signals. Connections across transformer encoders were considered in both models, aiming for more stable information exchange and better feature extraction. Later, Sun et al. [37] mainly focused on the tokenization process and developed a 2-D and 3-D convolution-integrated transformer that can express feature semantics more adequately. The more advanced Swin transformer was also employed together with the VGGNet regional feature extractor by Jamali and Mahdianpari [38], for classifying complex RS landscapes using the Sentinel-1, Sentinel-2, and LiDAR data, which is applicable to tasks of wetland mapping, monitoring, and so on. The specific center position encoding was initially introduced by Zhao et al. [39] in a hybrid convolutional transformer. Besides, Dong et al. [40] first explored the potential of ViT under the contrastive learning scheme for LULC classification using polarimetric SAR data. Xue et al. [18] and Roy et al. [41] have also made their first attempts to fuse HS and LiDAR data by means of modifying transformers, respectively.

Throughout the above-mentioned related work, although remarkable improvements evaluated by LULC classification

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:43:18 UTC from IEEE Xplore. Restrictions apply.

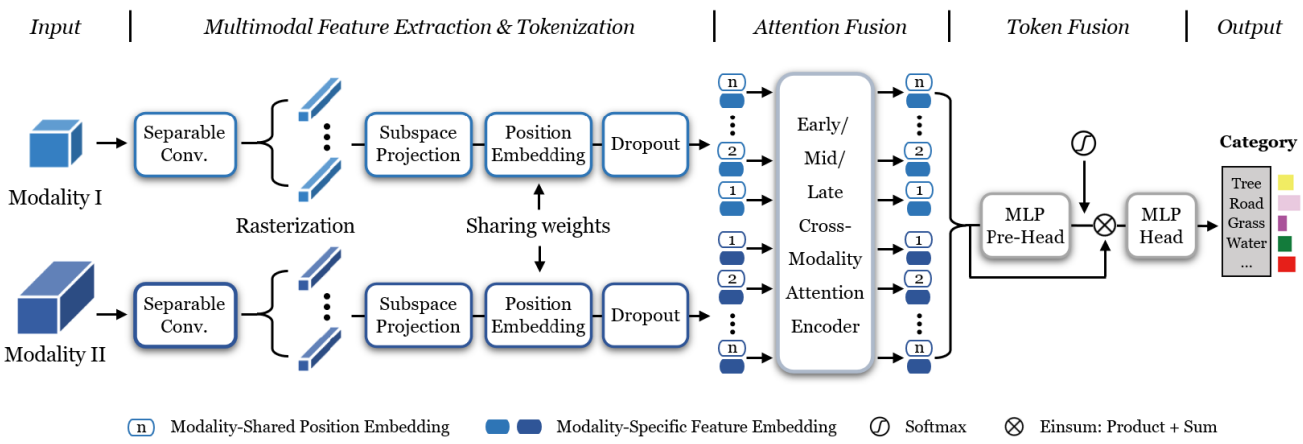


Fig. 3. Schematic workflow of the proposed ExViT method, which is primarily composed of three key constituent modules, i.e., separable convolution module, CMA module, and classification token fusion module.

performance have been made, most of which were still restricted to their exclusive designs of network architecture by following conventional encoding rules. Hence, we are in need of a more general framework that is modality-agnostic and suitable for more practical scenarios under multimodal and cross-modal settings.

III. Material and Methodology

In this section, we first give out a whole picture of our proposed deep framework that informs our design of three highlighted constituent modules. Then, we show the details of constructing our LULC classification network by considering the case of input data with two RS modalities. At last, we make a necessary analysis of the architecture of our method to illustrate its rationality.

A. Method Overview

The proposed method, namely ExViT, is a novel attempt to implement the attention mechanism in excavating the underlying relevant and complementary information among multimodal RS data. As depicted in Fig. 3, taking two-modality patch-based input as an example, our method adopts a dual-to-single hierarchical network architecture for effective feature extraction and fusion. There are three key modules employed in the proposed ExViT, i.e., separable convolution module, CMA module, and classification token fusion module, on the basis of ViT backbone, aiming to learn more expressive representation by comprehensive modeling of contextual and local discrepancies across modalities. By doing so, our ExViT offers a lightweight path to recognize LULC categories with more preferable performance and simultaneously enjoys promising extensibility.

B. Multimodal Feature Extraction and Tokenization

ViT-based models in the discipline of computer vision were originally invented for natural image classification. The setting of this task exhibits clear differences with the LULC classification based on remotely sensed images. To mention a few, pretraining on large-scale natural image datasets like ImageNet largely benefits the learning capacity of ViT, making it possible to attain breakthrough performance that is comparable to the dominant CNN-based models [42]. Also, the recognition of image-level semantics relies more on capturing global or regional salient patterns other than on detailed information mining among neighboring pixels, which may account for the fact that simple linear projection of non-overlapping patches scheme is enough to lead to the success of numerous ViT variants. It is worth noting that, however, the investigated LULC classification principally focuses on the categorical attribute according to each pixel for a given scene. Therefore, proper exploitation of spatial contextual information and consideration of network design are needed by taking the ground object characteristics and availability of limited training data into account.

Formally, given $\mathbf{x}^m \in \mathbb{R}^{H\times W\times d_m}, m = 1, 2$[1] collecting $N = H \times W$ pixels with d_m channels of mth modality observed image, our target is to predict the corresponding LULC mapping $\mathbf{y} \in \mathbb{R}^{N\times K}$ with each row $\mathbf{y}_i = [y_{i,1}, y_{i,2}, \ldots, y_{i,K}]$ denoting one-hot encoded representation, i.e., $y_{i,k=c} = 1, y_{i,k\neq c} = 0$ when ith pixel belongs to cth category. To better consider the spatial properties and thus ensure more realistic classification maps, we propose to build our model on patch-based input as the CNN-based model does. Consider input patch for ith pixel of each modality $\mathbf{x}_i \in \mathbb{R}^{P\times P\times d_m}$ where P denotes the spatial width, we explore an alternative network stem by means of the widely used separable convolution [43]. The adopted separable convolution module, as the middle part in Fig. 3 depicts, is designed by stacking L_1 blocks with each consisting of a *depthwise (DW) convolution* layer, i.e., 3×3 spatial convolutions operated independently on each channel of the last layer output, a *pointwise (PW) convolution* layer, i.e., 1×1 convolutions that integrate and map the channel's output by DW convolution into a new channel space, and a non-linear activation layer

[1]We consider RS data with two modalities here as the input of the proposed model, which is extensible to multimodal cases without loss of generality. For the sake of notation brevity, we will omit the superscript m without bringing ambiguity hereinafter and use a specific number instead if needed.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:43:18 UTC from IEEE Xplore. Restrictions apply.

using the Gaussian error linear units function [44], which is consistent with that used in the subsequent ViT encoders [42].

As the left part in Fig. 3 shows, similar to ViT which partitions an image into non-overlapping patches, we rasterize the output feature maps of the separable convolution module into P^2 vectors with each one representing a pixel within the patch. Those embedded features are then processed following the regular tokenization rule and get a series of 1-D tokens $\mathbf{z}_i \in \mathbb{R}^{P^2 \times d}$ that are ready for the next attention module. After that, we set linear layers for different modalities, hoping to project their extracted features into a common subspace. Note that this implicit feature alignment is necessary for better fusion of multimodal features in the next stages.

By combining the above-mentioned operations into a whole learnable function $g()$ and obeying conventional ViT rules [42], we then have the following tokenization form for ith patch:

$$\mathbf{z}_i = \mathrm{DP}([\mathbf{z}_{\mathrm{cls}}; g(\mathbf{x}_{i,1}); g(\mathbf{x}_{i,2}); \ldots; g(\mathbf{x}_{i,P^2})] + \mathbf{e}) \tag{1}$$

where $[;]$ denotes the concatenation along the first dimension, $\mathbf{z}_{\mathrm{cls}}$ is a special token prepended to this sequence so that its representation at the final layer can be passed to a classifier for classification tasks, $\mathbf{e} \in \mathbb{R}^{(P^2+1)\times d}$ is a learned positional embedding added to the tokens to retain positional information, and DP() denote the dropout layer to prevent our LULC classification network from overfitting issue [45]. To better align multimodal tokens and narrow the solution space, we consider sharing the position embedding $\mathbf{e}$ for the same patch of different modalities. In the following, we will see that the classification token $\mathbf{z}_{\mathrm{cls}}$ can be further pruned through our proposed attention- and decision-level feature fusion.

C. Multimodal Attention and Token Fusion

The pixelwise tokens of two modalities images are then parallelly passed into an encoder for further feature extraction. Similar to the concurrent related works, by revisiting regular ViT encoder architecture that includes multihead self-attention (MHSA), layer normalization (LN), and MLP layers [42], the core of which lies in the MHSA for each modality implemented by

$$\mathrm{MHSA}(\mathbf{w}|\mathbf{z}_i) = \mathrm{DP}([\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_A]\mathbf{w}^O). \tag{2}$$

In each head $\mathbf{h}_a$ out of A heads, the intramodality spatial correlation can be captured by conventional self-attention in the form of matrix inner product as

$$\begin{aligned}\mathbf{h}_a &= \mathrm{Attention}\left(\mathbf{z}_i\mathbf{w}_a^Q, \mathbf{z}_i\mathbf{w}_a^K, \mathbf{z}_i\mathbf{w}_a^V\right) \\ &= \mathrm{Softmax}\left(\frac{\left(\mathbf{z}_i\mathbf{w}_a^Q\right)\left(\mathbf{z}_i\mathbf{w}_a^K\right)^\top}{\sqrt{d}}\right)\left(\mathbf{z}_i\mathbf{w}_a^V\right).\end{aligned} \tag{3}$$

Note that only the parameters of FC layers $\{\mathbf{w}_a^Q, \mathbf{w}_a^K, \mathbf{w}_a^V\}$ and $\mathbf{w}^O$ need to be learned for projecting $\mathbf{z}_i$ into different representation subspaces of ath head and aggregating those multihead features, respectively.

Based on the above-mentioned contextual and local discrepancy information extraction within each modality, further consideration of modality interaction from feature and classification token fusion perspectives is worthwhile. The cross-modal attention mechanism has been gaining wide and increasing interest in the RS community recently [46], [47], including the DHViT proposed in the latest study [18]. However, the cross attention in DHViT only involves the utilization of querying one modality by the other one through the single use of classification token. Although it is able to abstract information from all tokens to some extent, a more general fusion strategy is needed in our proposed multimodal framework. Motivated by the fact that features extracted from shallow to deep layers tend to exhibit preferred information at different levels [30], we first propose attention fusion within ViT encoders. By straightforward concatenation of tokens from two modalities along the modality dimension $\tilde{\mathbf{z}}_i = [\mathbf{z}_i^1; \mathbf{z}_i^2]$, and assembling the building blocks mentioned earlier using residual connections for multimodal fused features, we can readily get the following forward function without breaking the self-attention rules as ViT obeys [42]:

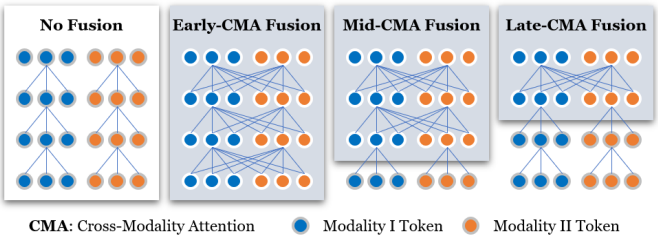


Fig. 4. Our proposed CMA module that adopts early-, mid-, and late-level attention fusion schemes.

$$\begin{aligned}\hat{\mathbf{z}}_i^{(l)} &= \mathrm{MHSA}\left(\mathrm{LN}\left(\tilde{\mathbf{z}}_i^{(l)}\right)\right) + \tilde{\mathbf{z}}_i^{(l)}, \ \tilde{\mathbf{z}}_i^{(l+1)} \\ &= \mathrm{MLP}\left(\mathrm{LN}\left(\hat{\mathbf{z}}_i^{(l)}\right)\right) + \hat{\mathbf{z}}_i^{(l)}\end{aligned} \tag{4}$$

where $0 \le l < L_2$ is the depth index indicating where the intermodality cross-attention begins. Since the token number and feature dimension remain unchanged throughout our encoder blocks, we can readily introduce a splitting scheme, e.g., $l = 0, \lfloor \alpha L_2 \rfloor, \lceil (1-\alpha) L_2 \rceil$ with $\alpha < 0.5$, to indicate early-, mid-, and late-level CMA fusion, respectively, as shown in Fig. 4, expecting a tunable cross-modal information exchange from low-level features to high-level semantics.

After full interactions between the classification tokens and other pixel-based tokens throughout L_2 consecutive encoders, the discriminative information can be progressively abstracted into the former ones. As usual, we can basically take out these tokens from two modalities [42] or perform further fusion by simple *or*, *concat*, and *mean* operations before feeding the fused result into the MLP head composed of LN and FC layers for final feature alignment and projection. However, we claim that the classification token is not always needed, since each token is handled equally while simply leaving out these tokens may result in inevitable information loss. Hence in our ExViT, we propose to fuse all tokens to get a final token for classification instead of introducing an extra $\mathbf{z}_{\mathrm{cls}}$ for each modality as (1) states. In specific, an MLP prehead similar to the MLP head is ably inserted with only $d \times 1$ parameters

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:43:18 UTC from IEEE Xplore. Restrictions apply.

being introduced. It acts as a squeeze operation to linearly project the final output tokens $\hat{\mathbf{z}}_i^{(L_2)}$ into 1-D vectors. Followed by a Softmax layer, these learned descriptors are expected to be expressive for corresponding tokens, and the weighted sum of them is finally computed and fed into the MLP head.

At last, the multiclass cross-entropy loss [25] based on multimodal training samples as follows is utilized to optimize the parameters of our network:

$$L(\mathbf{y}, \tilde{\mathbf{u}}|\mathbf{x}^1, \mathbf{x}^2) = -\frac{1}{|\Omega_{\text{train}}|} \sum_{i \in \Omega_{\text{train}}} \mathbf{y}_i \log(\tilde{\mathbf{u}}_i) \tag{5}$$

where $|\Omega_{\text{train}}|$ denotes the cardinality of the training index set and $\tilde{\mathbf{u}}_i$ represents the output unnormalized scores of ith pixel for each class.

D. Method Analysis

1) Separable Convolutional Stems Matter: Building on top of an RS image patch-based spatial ViT backbone, we make minimal architecture modifications to obtain the proposed ExViT method, which forms a general multimodal deep learning framework for LULC classification. In contrast to existing transformer-based methods that adopt either linear projection for patch embedding or pretrained VGGNet to extract regional features, we adopt grid feature output by separable convolutions within the local image patches. This is mainly aimed to learn an enriched and compact representation for each neighboring pixel by exploiting contextual information and therefore benefit the later feature embedding, self-modality attention and CMA. The PW convolution is also necessary for adaptively combining related information along the channel dimension, as the role of groupwise spectral embedding module plays in SpectralFormer [17], however avoiding using 3-D convolution that tends to be hard to train with limited labelings.

2) Why CMA Works?: Our employed CMA module affords not only flexibility in generalizing to the multimodal cases without introducing additional parameters but also interpretability in handling multimodal information exchange. More concretely, recalling the attention implementation in (3) and token concatenation in our cross-attention, let $\mathbf{q}_i^m = \mathbf{z}_i^m \mathbf{w}_a^Q, \mathbf{k}_i^m = \mathbf{z}_i^m \mathbf{w}_a^K, \mathbf{v}_i^m = \mathbf{z}_i^m \mathbf{w}_a^V$ denote the query, key, and value matrix of mth modality ith pixel, we have

$$\text{Softmax}\left(\frac{[\mathbf{q}_i^1; \mathbf{q}_i^2][\mathbf{k}_i^1; \mathbf{k}_i^2]^\top}{\sqrt{d}}\right)[\mathbf{v}_i^1; \mathbf{v}_i^2] = \left[\mathbf{s}_i^{1,1}\mathbf{v}_i^1 + \mathbf{s}_i^{1,2}\mathbf{v}_i^2; \mathbf{s}_i^{2,2}\mathbf{v}_i^2 + \mathbf{s}_i^{2,1}\mathbf{v}_i^1\right] \tag{6}$$

where $\mathbf{s}_i^{m_1,m_2}$ measures the similarity between m_1th and m_2th modality tokens. It is noteworthy that $\mathbf{s}_i^{1,2}$ and $\mathbf{s}_i^{2,1}$ have distinct normalization terms, thereby the second term of fused tokens can be regarded as complementary features for each modality by querying the other one, which considers all modalities more comprehensively than related works do [18]. In fact, we observe evidence that the discriminativeness of each modality's classification token improves drastically through this cross-attention manner. Moreover, since the patch size is commonly set small, albeit the computational and memory cost attention scales quadratically with respect to the token number, the resulting extra load is still worthwhile compared to evident performance gains.

3) How ExViT Handles Cross-Modal Learning?: As is often the case when a sufficient supply of multimodal data is not guaranteed in reality, caused by inevitable technical or sensing environment limitations under certain situations, we still expect the proposed DL framework to work for LULC classification. Fortunately, the settings of cross-modal learning commonly allow us with a limited amount of multimodal training sources [30]. Unlike existing cross-modal learning methods [31], [48], which directly extend the multimodal training samples $\{\mathbf{x}^1, \mathbf{x}^2\}$ by zero-padding the mth modality that is available in the testing phase $\{\mathbf{x}^m, \mathbf{0}\}$, we intend to introduce the idea of multitask learning for our ExViT by minimizing the following loss function [49]:

$$L_{\text{CML}} = \beta L(\mathbf{y}, \tilde{\mathbf{u}}|\mathbf{x}^1, \mathbf{x}^2) + \gamma L(\mathbf{y}, \tilde{\mathbf{u}}|\mathbf{x}^m, \mathbf{0}) \tag{7}$$

where $\beta, \gamma \geq 0$ denotes the tradeoff parameter to control the relative importance of a single-modality learning task as compared to the multimodal one. Through this manner, a flexible and interpretable joint optimization scheme is provided rather than simply augmenting the training set, aiming for better exploitation of cross-modal correspondence. Last but not least, the conventional technique adopted in [31] and [48] can also be included as a special case of ours by setting both β and γ to 1.

IV. Experiment and Discussion

In this section, we evaluate the proposed EvViT model by conducting LULC classification experiments on two benchmark RS datasets, i.e., the Houston2013 dataset and Berlin dataset, as shown in Fig. 5. We first briefly describe the details of the investigated multimodal data. Then, the evaluation metrics and implementation details are provided. After we carry out ablation studies mainly using the Houston2013 multimodal data to validate the effectiveness of employed modules, an extensive comparison with representative multimodal competitors and results analysis are finally presented to show the superiority of our method.

A. Datasets Description

1) Houston2013 HS-LiDAR Dataset: This dataset comprises an HS image and LiDAR-derived digital surface model (DSM) data, which were captured by the ITRES CASI-1500 imaging sensor in June 2012. The corresponding scene of both modalities covers the University of Houston campus and its surrounding urban area, TX, USA, sharing the same ground sampling distance (GSD) of 2.5 m. There are 349 $\times$ 1905 pixels appearing in this scene with the HS image consisting of 144 spectral bands covering 380–1050 nm region at 10 nm spectral resolution, and the co-registered LiDAR data containing only one channel that records height information. In terms of LULC classification, the studied scene is annotated with pixel-level labels of 15 meaningful classes. We follow the training and testing sets split prepared by the 2013 IEEE Geoscience and RS Society Data Fusion Contest,[2] which is

[2]http://www.classic.grss-ieee.org/community/technical-committees/data-fusion/2013-ieee-grss-data-fusion-contest/

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:43:18 UTC from IEEE Xplore. Restrictions apply.

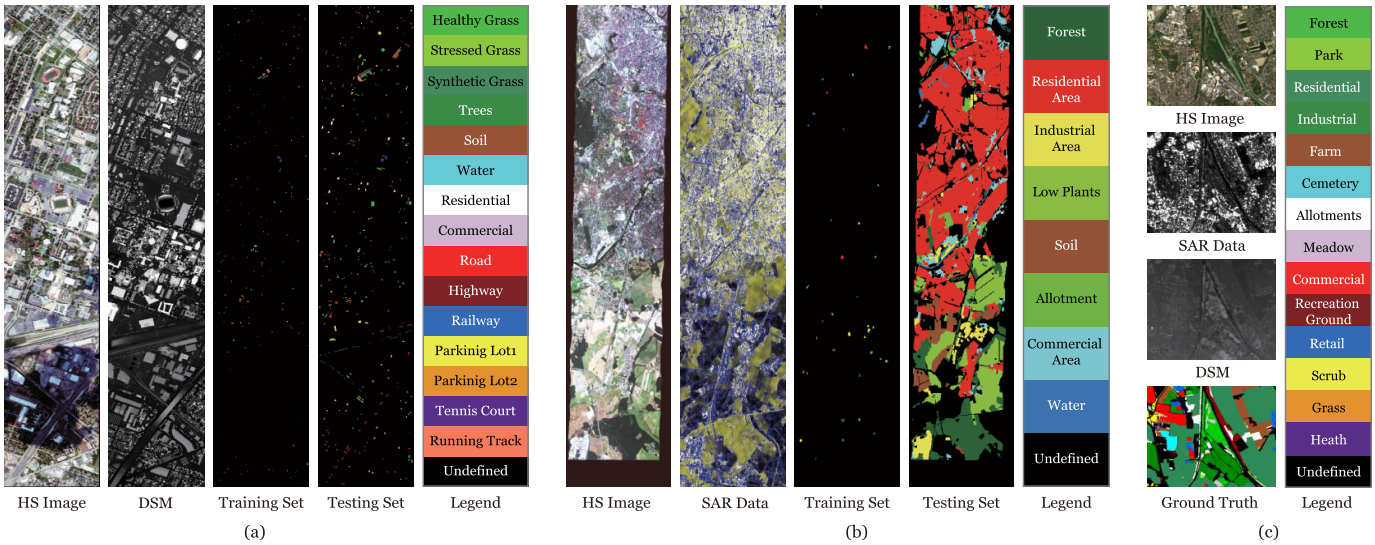


Fig. 5. Illustration of three investigated publicly available multimodal RS datasets. (a) Houston2013 dataset. (b) Berlin dataset. (c) Augsburg dataset.

TABLE I

DESCRIPTION OF THE INVESTIGATED HOUSTON2013 HS-LIDAR DATASET, INCLUDING THE TYPES OF GROUND OBJECTS AND THE CORRESPONDING NUMBERS OF TRAINING AND TESTING SAMPLES

Class No.	Class Name	Training Set	Testing Set
1	Healthy Grass	198	1053
2	Stressed Grass	190	1064
3	Synthetic Grass	192	505
4	Tree	188	1056
5	Soil	186	1056
6	Water	182	143
7	Residential	196	1072
8	Commercial	191	1053
9	Road	193	1059
10	Highway	191	1036
11	Railway	181	1054
12	Parking Lot1	192	1041
13	Parking Lot2	184	285
14	Tennis Court	181	247
15	Running Track	187	473
Total	-	2832	12197

TABLE II

DESCRIPTION OF THE INVESTIGATED BERLIN HS-SAR DATASET, INCLUDING THE TYPES OF GROUND OBJECTS AND THE CORRESPONDING NUMBERS OF TRAINING AND TESTING SAMPLES

Class No.	Class Name	Training Set	Testing Set
1	Forest	443	54511
2	Residential Area	423	268219
3	Industrial Area	499	19067
4	Low Plants	376	58906
5	Soil	331	17095
6	Allotment	280	13025
7	Commercial Area	298	24526
8	Water	170	6502
Total	-	2820	461851

TABLE III

DESCRIPTION OF THE INVESTIGATED AUGSBURG HS-SAR-LIDAR DATASET, INCLUDING THE TYPES OF GROUND OBJECTS AND THE CORRESPONDING NUMBERS OF TRAINING AND TESTING SAMPLES

Class No.	Class Name	Training Set	Testing Set
1	Forest	1235	4936
2	Park	55	220
3	Residential	8704	34812
4	Industrial	2487	9947
5	Farm	1456	5821
6	Cemetery	321	1284
7	Allotments	866	3460
8	Meadow	256	1023
9	Commercial	950	3797
10	Recreation Ground	598	2389
11	Retail	340	1358
12	Scrub	236	941
13	Grass	72	285
14	Heath	36	141
Total	-	17612	70414

well separated as shown in Fig. 5(a). The detailed statistics including class names and sample sizes are summarized in Table I.

2) Berlin HS-SAR Dataset: The Berlin dataset we use is comprised of a simulated EnMAP HS image and SAR data that comes from Sentinel-1 payload, covering the same subset describing Berlin city and its neighboring rural area. The original 30-m-GSD HS image with 797 × 220 pixels is a spatially and spectrally simulated product derived from the 9 m airborne HyMap data, which were acquired in August 2009 and made freely available online [50].[3] The SAR data with 13.89 m GSD were downloaded from ESA and preprocessed with the help of ESA toolbox SNAP.[4] There are 1723 × 476 pixels contained in this scene with the spatially resampled HS image consisting of 244 spectral bands in the wavelength range of 400–2500 nm, and the co-registered SAR image involving four channels derived from the covariance matrix of the dual-polarimetric (VH-VV) single look complex product. The ground reference map provided by OpenStreetMap (OSM) and refined using Google Earth is adopted [51], which contains eight categories as details listed in Table II.

[3]http://doi.org/10.5880/enmap.2016.002

[4]https://step.esa.int/main/toolboxes/snap/

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:43:18 UTC from IEEE Xplore. Restrictions apply.

TABLE IV

QUANTITATIVE RESULTS OF COMPARED METHODS IN TERMS OF OA, AA, κ, CLASSWISE ACCURACY, AND CORRESPONDING STANDARD DEVIATION σ ON *Houston2013 HS-LiDAR* DATASET, IN WHICH THE BEST ONES ARE HIGHLIGHTED IN BLUE

Method	Stack-NN	Stack-RF	CoSpace	LeMA	Co-CNN	CCR-Net	Co-ViT	Co-SF	DHViT	ExViT
Healthy Grass	82.62	82.53	81.96	81.86	83.10	83.00	76.19	76.19	81.01	91.55
Stressed Grass	81.77	83.36	83.27	83.08	84.87	84.87	76.42	75.47	85.15	85.15
Synthetic Grass	100.00	98.42	100.00	100.00	99.80	100.00	100.00	100.00	93.47	98.61
Trees	91.48	91.48	94.22	94.79	92.42	92.14	97.14	92.38	80.40	98.61
Soil	96.50	96.88	99.34	99.34	99.24	99.81	97.14	96.19	99.53	100.00
Water	95.10	98.60	99.30	99.30	95.80	95.80	100.00	85.71	95.80	98.60
Residential	77.99	80.04	81.44	88.99	95.24	95.34	86.92	93.46	92.16	88.90
Commercial	66.48	43.02	66.10	74.26	81.86	81.39	73.33	68.57	92.21	93.35
Road	70.07	69.97	69.97	73.84	85.08	84.14	80.00	79.05	81.78	89.52
Highway	58.01	45.85	48.94	72.20	61.10	63.22	50.49	48.54	68.73	65.54
Railway	74.48	74.38	88.61	82.26	83.87	90.32	82.86	82.86	79.60	95.83
Parking Lot 1	76.66	67.82	88.57	90.30	91.26	93.08	87.50	81.73	94.81	90.39
Parking Lot 2	57.54	64.56	68.07	67.37	88.77	88.42	92.86	78.57	88.42	90.53
Tennis Court	100.00	100.00	100.00	100.00	91.09	96.36	100.00	95.83	100.00	99.60
Running Track	98.52	97.67	98.31	98.10	98.73	99.37	97.87	97.87	71.25	89.85
OA (%)	79.57	76.19	82.14	85.42	87.23	88.15	83.18	81.45	85.82	90.62
AA (%)	81.81	79.64	84.54	87.05	88.82	89.82	86.58	83.50	86.95	91.76
σ	15.16	18.68	15.45	11.48	9.80	9.65	13.83	13.59	9.67	8.64
κ	0.7793	0.7440	0.8062	0.8417	0.8619	0.8719	0.8182	0.7997	0.8461	0.8983

3) Augsburg HS-SAR-LiDAR Dataset: The third dataset comes as a part of a multimodal benchmark called MDAS that was newly provided in [52]. It comprises an EeteS-simulated EnMAP HS image [53], a DLR 3K DSM, and Sentinel-1 SAR data, covering a subregion of 10 km^2 of the city of Augsburg, Germany. To enhance the variability of the investigated multimodal datasets, a 2.2 m GSD HySpex imagery was used as the source data to produce more realistic EnMAP HS image with 242 spectral bands, and only the VH and VV channels of a dual-Pol SAR data were used, which are different from the above-mentioned Berlin dataset. All the image data were spatially adjusted to a 10 m GSD, resulting in a total of 300 $\times$ 360 pixels as shown in Fig. 5(c). The pixelwise LULC cover references containing 14 classes were generated by manually labeling based on OSM, as the detailed training and testing split information was summarized in Table III by following settings in [52].

The first two datasets have been made ready-to-use online[5] [26], and the last one can be found from this work[6] [52].

B. Experimental Setup

1) Evaluation Metrics: We mainly adopt three widely used metrics, i.e., overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ), and standard deviation (σ) of classwise accuracy, to quantitatively evaluate the performance of LULC classification methods. The values of these metrics range from 0 to 1 and are generally the larger the better. Specifically, by collecting all pixelwise predictions and ground-truth labels, OA, AA, and κ can be computed by

$$\mathrm{OA} = \frac{N_c}{N_a}, \quad \mathrm{AA} = \frac{1}{K}\sum_{j=1}^{K}\frac{N_c^j}{N_a^j}, \quad \kappa = \frac{\mathrm{OA} - P_e}{1 - P_e} \tag{8}$$

where N_c and N_a are the numbers of correctly classified and total test samples, and N_c^j and N_a^j are those numbers for specific jth class. The definition of κ considers the class imbalance factor by the means of the hypothetical probability of chance agreement P_e, which follows [54]:

$$P_e = \frac{\sum_{j=1}^{K} N_a^j \times N_p^j}{N_a \times N_a} \tag{9}$$

where N_p^j is the number of predicted samples for jth class.

2) Multimodal Classification Methods for Comparison: We select several state-of-the-art (SOTA) multimodal algorithms to validate the effectiveness of the proposed method from both visual and quantitative perspectives. These methods are representative of four main groups: 1) classic classifiers based on direct stacking of multimodal data: nearest neighbors (Stack-NN), random forest (Stack-RF) [55]; 2) joint feature learning algorithms: CoSpace[7] [15], LeMA[8] [56]; 3) multimodal CNN-based models: Co-CNN[9] [57], CCR-Net[10] [16]; and 4) transformer-based networks: Co-ViT,[11] Co-SF,[12] and DHViT [18], most of which have made their code publically available.

3) Implementation Details: We build our ExViT model on the basis of the PyTorch machine learning framework using a PC with 64-GB memory and an Nvidia GeForce GTX 3080 GPU card. Concretely, we adopt the Adam optimizer to update the network parameters with a batch size of 64. The learning rate is initially set as 5e^{-4} and then decayed by 0.9 every 20 steps, and weight decay is parameterized by 0 and 5e^{-4}, respectively. The image patches with 13 $\times$ 13 pixels are fed into the separable convolution module with $L_1 = 3$ consecutive blocks. The settings of the transformer encoder are as

[5]https://github.com/danfenghong/ISPRS_S2FL

[6]https://mediatum.ub.tum.de/1657312

[7]https://github.com/danfenghong/IEEE_TGRS_CoSpace

[8]https://github.com/danfenghong/ISPRS_LeMA

[9]https://github.com/RenlongHang/Coupled-CNNs

[10]https://github.com/danfenghong/IEEE_TGRS_CCR-Net

[11]We made appropriate modifications on vanilla ViT and SpectralFormer to obtain their multimodal versions for a fair comparison.

[12]https://github.com/danfenghong/IEEE_TGRS_SpectralFormer

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:43:18 UTC from IEEE Xplore. Restrictions apply.

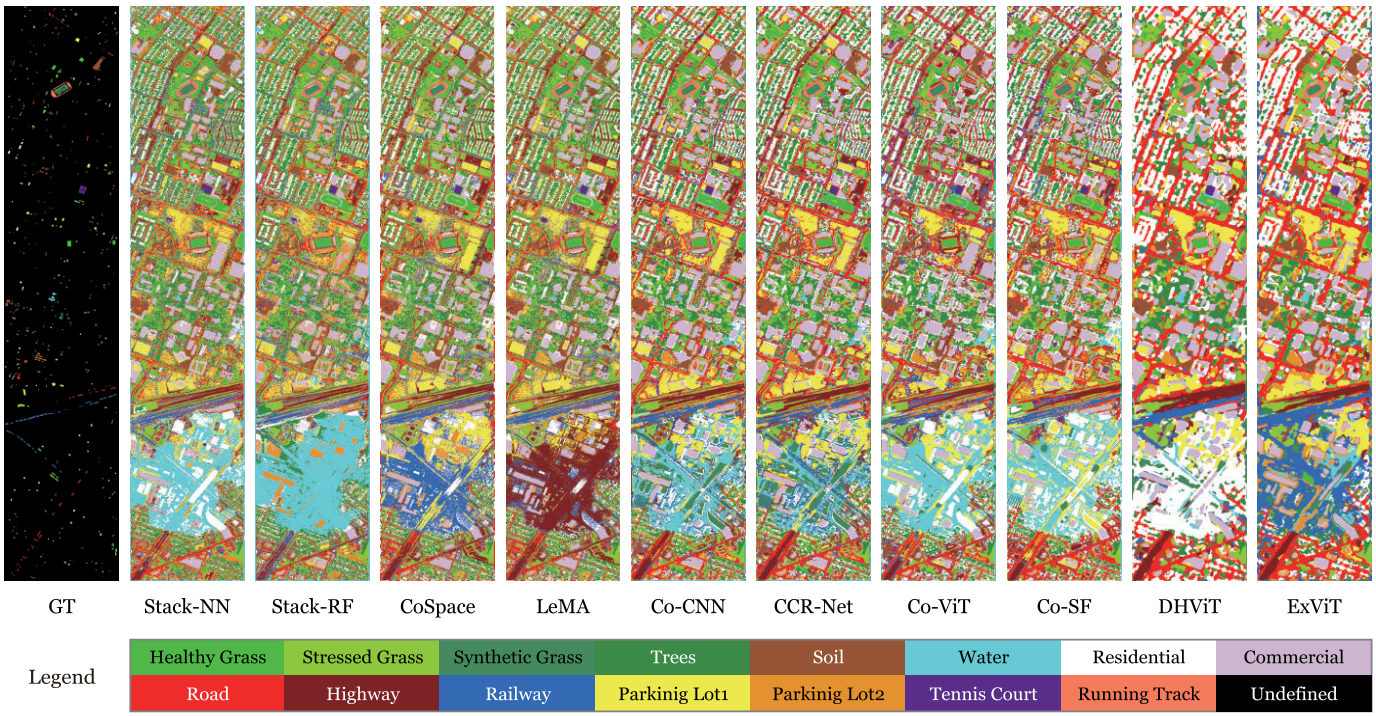


Fig. 6. LULC classification maps obtained by compared methods using *Houston2013 HS-LiDAR* multimodal data.

TABLE V

QUANTITATIVE RESULTS OF DIFFERENT COMPARED METHODS IN TERMS OF OA, AA, κ, CLASSWISE ACCURACY, AND CORRESPONDING STANDARD DEVIATION σ ON *Berlin HS-SAR* DATASET, IN WHICH THE BEST ONES ARE HIGHLIGHTED IN BLUE

Method	Stack-NN	Stack-RF	CoSpace	LeMA	Co-CNN	CCR-Net	Co-ViT	Co-SF	DHViT	ExViT
Forest	73.57	79.31	85.09	85.11	84.09	85.93	67.12	54.15	72.41	70.44
Residential Area	42.93	50.46	61.60	64.84	68.48	71.04	71.16	73.36	71.11	72.48
Industrial Area	38.72	41.64	51.18	48.94	49.09	41.80	40.45	47.02	29.43	46.95
Low Plants	69.23	65.82	75.44	80.04	79.43	82.62	89.54	84.57	88.69	81.71
Soil	81.01	66.94	82.50	80.66	81.25	85.10	80.22	92.67	80.58	91.35
Allotment	62.89	61.01	54.66	54.07	50.68	63.02	47.12	51.57	66.56	79.16
Commercial Area	36.83	35.84	28.81	27.40	26.16	29.23	36.93	7.30	35.22	36.35
Water	67.13	66.98	60.78	57.75	59.52	68.78	76.62	72.52	73.50	79.81
OA (%)	51.71	55.82	64.53	66.71	68.51	69.85	69.66	68.01	70.14	71.44
AA (%)	59.04	58.50	62.51	62.35	62.34	66.99	67.12	60.39	64.69	69.78
σ	17.09	14.63	18.54	19.51	19.96	20.76	19.66	26.88	21.12	18.68
κ	0.3838	0.4208	0.5093	0.5312	0.5476	0.5716	0.5639	0.5353	0.5735	0.5934

follows: hidden layer width of 64, encoder depth $L_2 = 6$ (half of 12 that is commonly used in natural image classification), head number of 4, and a dropout rate of 0.1. We empirically observe that 500 epochs training is able to find a converged status with satisfying results on both aforementioned two datasets.

C. Comparison Result and Analysis

1) Houston2013 HS-LiDAR Dataset: We first present the classification results of different competitors using HS and LiDAR data on the Houston2013 dataset. The quantitative results in terms of three metrics, i.e., OA, AA, and κ, as well as the classwise accuracy, are summarized in Table IV, while the corresponding classification maps and mapping legend are shown in Fig. 6.

Generally, the primitive NN method based on stacked multimodal input can provide an acceptable classification result with near 80% OA, which is better than stack-RF. More advanced conventional methods such as CoSpace and LeMA further increase the performance to a competitive level, but still 2% behind the deep method Co-CNN. Besides, the other CNN-based method CCR-Net surprisingly achieves the best performance in seven classes owing to its elaborate design of an effective cross-channel reconstruction module. The transformer-based competitors, i.e., Co-ViT, Co-SF, and DHViT, do not perform that well with their OAs less than that of CNN-based methods, while our ExViT can observably perform better than all the competitors in terms of all three metrics. It is also noteworthy that the lowest standard deviation of our method, to a great extent, demonstrates its overall stable performance in classifying different categories.

We can draw similar evidence from the visualized maps in Fig. 6. The overall zone partitions in the top half part by all compared methods clearly exhibit physically meaningful boundaries. However, the shadow of a cloud that appears in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:43:18 UTC from IEEE Xplore. Restrictions apply.

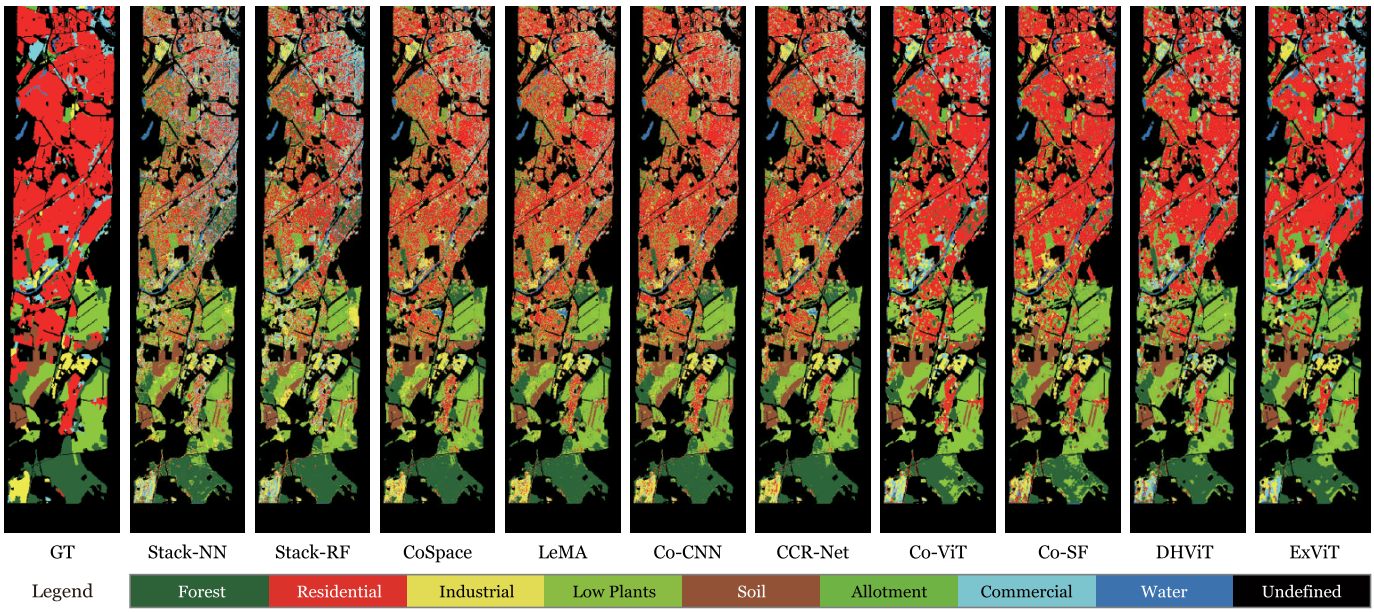


Fig. 7. LULC classification maps obtained by compared methods using *Berlin HS-SAR* multimodal data.

TABLE VI

QUANTITATIVE RESULTS OF COMPARED METHODS IN TERMS OF OA, AA, κ, CLASSWISE ACCURACY, AND CORRESPONDING STANDARD DEVIATION σ ON *Augsburg HS-SAR-LiDAR* DATASET, IN WHICH THE BEST ONES ARE HIGHLIGHTED IN BLUE

Method	Stack-NN	Stack-RF	CoSpace	LeMA	Co-CNN	CCR-Net	Co-ViT	Co-SF	DHViT	ExViT
Forest	69.51	79.76	78.85	84.06	95.36	94.96	96.29	95.06	95.75	96.13
Park	3.64	5.91	5.91	7.73	45.45	47.73	70.45	69.09	74.55	72.27
Residential	70.43	93.34	95.12	95.98	99.43	99.37	97.76	98.22	98.21	98.64
Industrial	50.59	59.80	61.98	63.45	93.50	98.54	95.42	96.07	96.92	98.60
Farm	79.57	87.27	87.89	88.87	92.44	97.37	96.34	93.92	94.73	95.52
Cemetery	32.01	63.01	60.98	72.35	97.27	98.68	95.17	98.29	99.45	99.53
Allotments	31.79	25.14	20.49	25.87	85.20	84.62	91.24	88.76	88.18	91.65
Meadow	50.15	53.96	54.35	58.06	84.75	85.73	80.55	82.21	82.11	88.86
Commercial	34.76	40.24	40.45	45.75	91.94	93.92	92.07	94.79	94.23	96.21
Recreation Ground	41.36	53.70	57.14	63.42	96.23	94.52	97.03	94.14	97.87	97.70
Retail	32.99	33.14	42.56	38.95	83.51	78.20	90.13	92.78	92.27	93.45
Scrub	21.36	27.84	39.85	42.51	87.78	79.91	90.12	85.76	85.76	91.07
Grass	9.47	10.88	4.56	9.82	53.33	61.40	62.11	65.96	76.49	76.14
Heath	19.15	29.08	33.33	36.17	76.60	75.18	80.14	92.91	87.94	88.65
OA (%)	60.58	75.69	77.06	79.16	95.40	96.35	95.75	95.82	96.17	97.17
AA (%)	39.06	47.36	48.82	52.36	84.49	85.01	88.20	89.14	90.32	91.74
σ	22.89	27.35	27.77	27.74	16.20	14.69	10.93	10.21	8.10	8.27
κ	0.4566	0.6341	0.6538	0.6864	0.9350	0.9487	0.9406	0.9415	0.9465	0.9606

the bottom area causes difficulty in recognizing ground objects within it. There exist evident differences in the obtained classification maps beneath a nearly horizontally oriented *Highway*. For example, CoSpace, LeMA, and DHViT tend to classify the majority of this area into the classes of *Railway*, *Highway*, and *Residential*, respectively, which do not quite meet with real situations. The estimation of our proposed ExViT shows clear patterns of ground objects that are more convincible. The other methods are prone to detect the wide existence of *Water*, possibly due to the fact that there still remains some water vapor effect.

2) Berlin HS-SAR Dataset: We conduct more experiments for comparison on the second Berlin dataset using HS and SAR data. Similarly, the quantitative and visual classification results are, respectively, shown in Table V and Fig. 7.

We can basically observe from the relatively lower performance in the table that this is a more challenging dataset. The gap between naive stack-NN/RF and the other advanced classifiers increases more evidently than that in the previous Houston2013 dataset, especially in terms of OA. The joint feature learning models, i.e., CoSpace and LeMA, show a much better estimation of the *Residential Area* that consists of a large proportion of this scene with red color. Unsurprisingly, there arises a significant performance increase from conventional to deep-learning-based methods. Among all these competing methods, our ExViT attains the best results in terms of OA, AA, and κ as expected, which are 1%–2% better than the second-best ones achieved by DHViT and Co-ViT, respectively. It is also the visual results of our ExViT that is most close to the GT. We note that, from the perspective of classwise accuracy, our method also outperforms the SOTA CCR-Net and DHViT substantially for most classes, showing its ability in identifying a wide range of LULC attributes by organically linking and mining multimodal information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:43:18 UTC from IEEE Xplore. Restrictions apply.

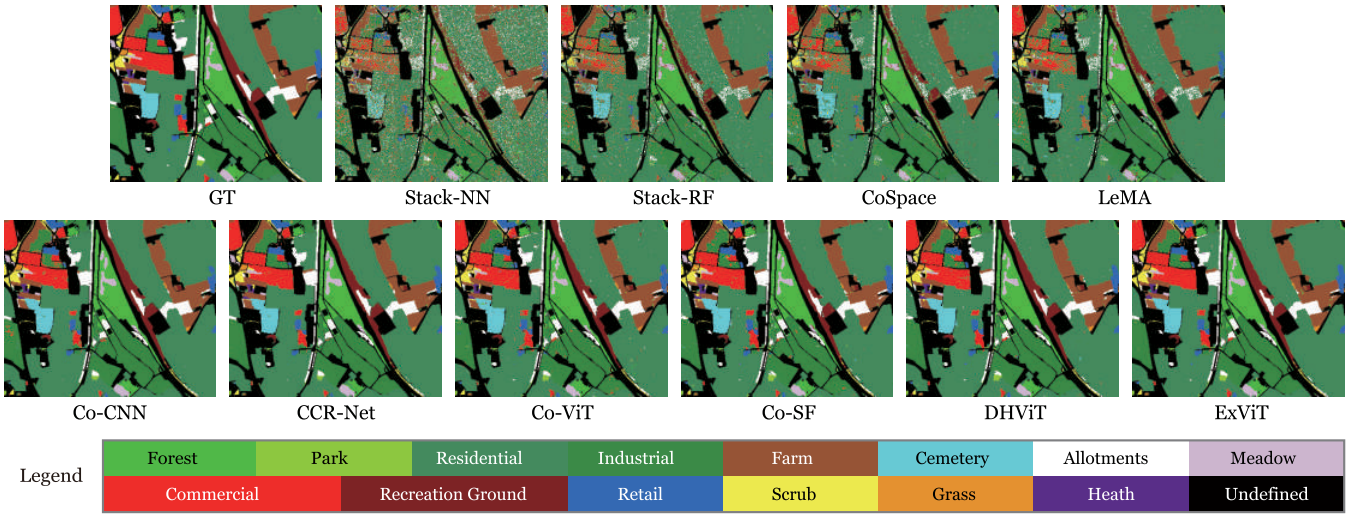


Fig. 8. LULC classification maps obtained by compared methods using *Augsburg HS-SAR-LiDAR* multimodal data.

TABLE VII

QUANTITATIVE RESULTS OF ABLATION STUDY ON EARLY-STAGE DW AND PW CONVOLUTIONS FOR DIFFERENT MODALITIES

Dataset	Houston2013					
Modality	HS			LiDAR		
Backbone	ViT	DViT	SViT	ViT	DViT	SViT
DW Conv.	✗	✓	✓	✗	✓	✓
PW Conv.	✗	✗	✓	✗	✗	✓
OA (%)	74.69	80.07	84.91	48.27	39.42	56.05
AA (%)	76.47	82.67	86.71	47.22	37.19	53.58
κ	0.7263	0.7840	0.8365	0.4403	0.3465	0.5235

TABLE VIII

QUANTITATIVE RESULTS OF ABLATION STUDY ON CMA FUSION AT DIFFERENT LEVELS

Attention	No Fusion	Early-CMA	Mid-CMA	Late-CMA
OA (%)	88.73	88.87	89.37	90.62
AA (%)	90.28	90.40	90.92	91.76
κ	0.8780	0.8794	0.8849	0.8983

TABLE IX

QUANTITATIVE RESULTS OF ABLATION STUDY ON VARYING CLASSIFICATION (CLS) TOKEN FUSION STRATEGIES

CLS Token	HS*	LiDAR*	Concate	Mean	Proposed
OA (%)	86.69	88.25	88.66	89.17	90.62
AA (%)	89.10	89.75	90.93	90.74	91.76
κ	0.8561	0.8728	0.8773	0.8827	0.8983

3) Augsburg HS-SAR-LiDAR Dataset: To further verify the expandability of our ExViT as a general multimodal framework, in this section, we conduct LULC classification with input of three heterogeneous modalities. Table VI and Fig. 8 summarize the quantitative classification and visual mapping results, respectively.

Different from the results in the above-mentioned two datasets, deep models drastically increase the classification accuracy from the level around 80% to above 95%. This phenomenon is partially due to the fact that the training set in this dataset is not as well separated as those in the former ones. We can also draw evidence from the speckle noise-like patterns in the mapping results of conventional methods, which typically focus on pixelwise feature learning. What is more interesting, RS transformer methods agree to attain much more better AA than that of CNN-based ones. This could possibly be accounted by the sufficient training samples. It is undoubtable to see that our ExViT achieves the best classification performance among all the competitors.

D. Ablation Study

1) Early-Stage Separable Convolution: We first validate the necessity of adopting separable convolution at the very beginning of the feature extraction stage. To better investigate the effect of such design on different modalities, we design experiments using spatial ViT backbone to learn from single-modality input. The classification results under different network settings are summarized in Table VII. From the table, we can observe that the ViT structured network is able to capture the advantages of using HS data other than LiDAR for the LULC classification. More specifically, the DW convolution and PW convolution both help the subsequent ViT see better from the HS data. However, the case for LiDAR differs, using DW convolution only causes significant decreases in all three metrics, but the joint use of these convolutions brings accuracies back. It could be accounted that the DW convolution tends to smooth the spatial patterns and eliminate the discrepant information conveyed by single-channel LiDAR data, while the PW convolution can enlarge the representation power by feature fusion across channels, thus benefiting the pixelwise tokenization.

2) CMA Fusion: To explore the performance gains brought by CMA fusion at different levels, we mainly consider three settings as early, mid, and late cases by fixing the settings of other parts. The first column in Table VIII refers to the result of two separate single-modality ViTs without attention fusion, which can indicate the effectiveness of the proposed feature

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:43:18 UTC from IEEE Xplore. Restrictions apply.

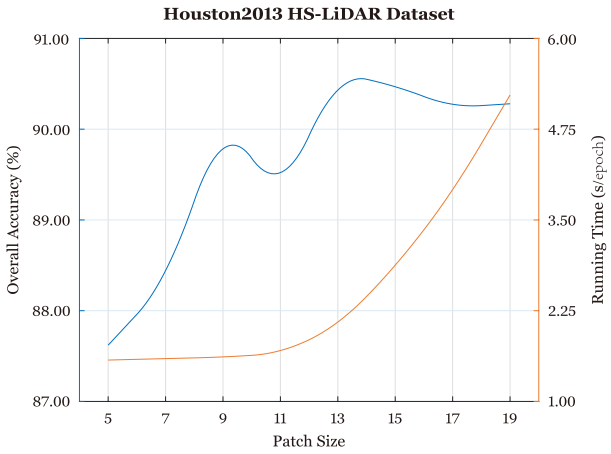


Fig. 9. Ablation study results in terms of OA and running time per epoch on varying sizes of the image patch.

extraction and token fusion from the other side by comparing it with Table VII. The performance of such a baseline attains an evident improvement after adding the attention fusion module. Specifically, fusing the multimodal tokens at late stage with CMA tends to generate better performance than its counterparts fusing at earlier stages. This could be explained by that the high-level features of different modalities abstracted by ViT encoders generally exhibit clear semantics, thus tending to yield more accurate fusion results.

3) Classification Token Fusion: We then study the discriminative ability of the classification token generated in various ways. In Table IX, we first report the results using conventional classification tokens of a single modality in the left four columns, whereas those marked with * denote the cases with our proposed late-stage CMA fusion. As expected, HS* and LiDAR* achieve similar improved results across all metrics, which verifies the successful information exchange through the cross-query on the other modality as (6) indicates. Given such a benefit, we further consider the integration of classification tokens of both modalities. Two kinds of vanilla fusion ways (concatenation and mean) are investigated in comparison with the proposed all-tokens fusion. It can be easily observed that further improvement in accuracy can be reached by averaging two classification tokens compared to their concatenation, while the proposed one performs the best with an evident improvement of around 1.5% OA.

4) Size of Image Patch: Although conventional ViT has earned a reputation for its ability in capturing long-range dependences, it is still worthwhile to analyze the effect by varying the input patch sizes of the multimodal RS image. We draw the ablation study results in terms of OA and running time in Fig. 9. With the increase in patch size P, the plot in blue first shows a two-stage dramatic rise and then runs stably after reaching the size of 17, which is relatively large as compared to CNN-based competitors [16]. The complex distribution of ground objects in RS scenes could possibly account for this phenomenon, especially under spaceborne sensing situations with large GSDs. Considering that the computational burden keeps increasing, we therefore utilize the patches of 13×13 in the following experiments. We can

TABLE X
QUANTITATIVE COMPARISON OF MODEL COMPLEXITY IN TERMS OF THE NUMBER OF PARAMETERS AND FLOPS

Method	CCR-Net	Co-ViT	Co-SF	DHViT	ExViT
Parameters (M)	0.20	0.22	0.28	0.48	0.23
FLOPs (G)	1.64	2.96	4.67	7.32	3.51

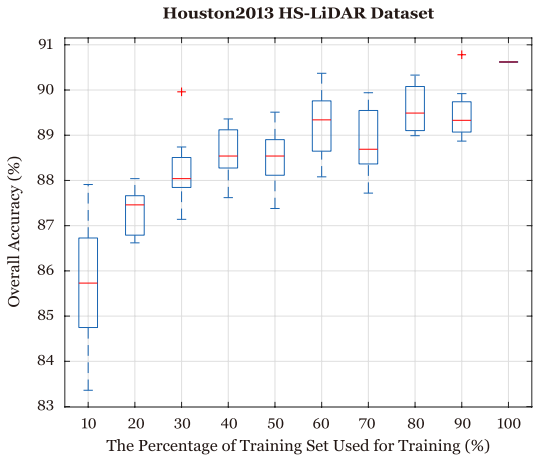


Fig. 10. Ablation study results in terms of OA on varying sizes of the training set.

also notice that the OAs at $P \geq 7$ are able to surpass that of CCR-Net in Table IV, which indicates the effectivity of our proposed cross-attention even when processing within a relatively local perception field.

5) Size of Training Set: Apart from the above-mentioned ablation study on key components in our network, we also investigate the issue of how our method relies on training data volume. We conduct ten rounds of repeated experiments using randomly selected training samples from the standard training set with a proportion from 10% to 90%. In Fig. 10, we draw a boxplot that summarizes the statistical characteristics of the obtained OAs. From the figure, an obvious trend shows that starting from OA around 86% using only 10% samples of the training set, the averaged OA then shows approximately constant in accordance with the increasing training data size. In fact, with only around 60%–80% data being available, our method can stably achieve a satisfying performance with 89%+ OA, which is already able to surpass that of the SOTA CCR-Net from Table IV (88.15%). This not only demonstrates the stability of our proposed network but also verifies its efficiency in fully mining supervised information with less training data.

6) Model Complexity: In practical cases, the users often care about the efficiency of the deep models, which typically consists of the cost from time and space perspectives. We select the number of learnable parameters and the floating point operations (FLOPs) in models to make a quantitative comparison between deep competitors. From the results summarized in Table X, we can draw several conclusions as follows. The CNN-based model CCR-Net enjoys the lowest complexity in terms all these two indicators, thanks to the weights sharing mechanism and spatial pooling operations.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:43:18 UTC from IEEE Xplore. Restrictions apply.

TABLE XI
QUANTITATIVE RESULTS OF ABLATION STUDY ON CROSS-MODAL MULTITASK LEARNING ABILITY

Modality	HS	LiDAR	HS†	LiDAR†	HS††	LiDAR††
(β, γ)	$(0, 1)$	$(0, 1)$	$(1, 1)$	$(1, 1)$	$(1, 2)$	$(1, 0.1)$
OA (%)	85.39	60.04	85.78	60.75	87.18	61.27
AA (%)	87.71	60.65	88.1	60.65	88.95	61.75
κ	0.8415	0.5676	0.8457	0.5746	0.861	0.5804

The parameters of Co-SF are slightly more than that of Co-ViT, while the former one involves more spectralwise attention computations between hundreds of bands. It has been observed in [18] that DHViT inevitably brought more parameters and computations due to its deeper architecture and hierarchical feature interactions, respectively. Note that our ExViT provides an acceptable complexity level in the middle of the five to strike a good balance between efficiency and effectiveness.

7) Cross-Modal Multitask Learning Ability: Another issue needs to be considered is the cross-modal learning ability. Due to the absence of some specific modalities (like spaceborne HS or airborne high-resolution LiDAR) in practical scenarios, it is crucial to investigate whether our proposed framework can still generate a stable classification performance. We fix the network architecture based on the above-mentioned ablation analysis and then assess its cross-modal learning ability by equivalently varying the multitask learning weight β and γ in (7), the results of which are summarized in Table XI. The first two columns correspond to the results when either LiDAR or HS modality is missing in testing, respectively, with β equals 0, which can be regarded as baselines by straightforwardly transferring network learned from bimodal training to single-modal testing. The next two columns with † refer to the existing cross-modal learning manner that directly uses bimodal and zero-padded single-modal training data, i.e., $\beta = \gamma = 1$ in our framework. Without laborious parameter tuning, by setting γ to 2 and 0.1, respectively, the best cross-modal classification performance can be achieved by our framework as the last two columns with †† show. These results also indicate that users are suggested to weigh the multitask learning objective more on the multimodal term to compensate for the large distribution gap when only the less informative modality is provided for the cross-modal inference in use.

V. CONCLUSION AND OUTLOOK

In this article, we explored a novel multimodal transformer-based framework, termed ExViT, to effectively tackle the practical task of LULC classification. To facilitate multimodal feature extraction and tokenization, our ExViT introduces the separable convolutional stems followed by subspace projections and shared position embedding layers, aiming for simultaneous exploitation of modality-specific information (from spatial to channel, from local to global) and intermodality relation (feature and positional alignment). The generic combination of self- and CMA fusion at different levels and classification token fusion prehead further empowers the spatial ViT, as the backbone of our ExViT, to take full advantage of its ability in modeling long-term correlation across modalities. As a whole, the superiority of the proposed ExViT lies in not only its uncluttered architecture with promising expansibility but also the extensive ablation studies and comparison experiments that show the SOTA performance.

In the future, challenges specific to developing more robust attention-based multimodal DL models against vulnerable modal-incomplete Earth observations, remotely sensed image segmentation methods that are able to interpret and react with high efficiency under practical cases with complex geospatial semantics, as well as building more multiscene multimodal RS datasets with reliable and separable ground labelings are worth our efforts.

REFERENCES

[1] L. Gómez-Chova, D. Tuia, G. Moser, and G. Camps-Valls, "Multimodal classification of remote sensing images: A review and future directions," *Proc. IEEE*, vol. 103, no. 9, pp. 1560–1584, Sep. 2015.

[2] S. Kogou et al., "From remote sensing and machine learning to the history of the silk road: Large scale material identification on wall paintings," *Sci. Rep.*, vol. 10, no. 1, pp. 1–14, Nov. 2020.

[3] J. Yao, D. Hong, L. Xu, D. Meng, J. Chanussot, and Z. Xu, "Sparsity-enhanced convolutional decomposition: A novel tensor-based paradigm for blind hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5505014, doi: 10.1109/TGRS.2021.3069845.

[4] X. Zhang, M. Pazner, and N. Duke, "Lithologic and mineral information extraction for gold exploration using ASTER data in the south chocolate mountains (California)," *ISPRS J. Photogramm. Remote Sens.*, vol. 62, no. 4, pp. 271–282, Sep. 2007.

[5] G. Chen, Q. Weng, G. J. Hay, and Y. He, "Geographic object-based image analysis (GEOBIA): Emerging trends and future opportunities," *GISci. Remote Sens.*, vol. 55, no. 2, pp. 159–182, Mar. 2018.

[6] R. Hang, Q. Liu, and Z. Li, "Spectral super-resolution network guided by intrinsic properties of hyperspectral imagery," *IEEE Trans. Image Process.*, vol. 30, pp. 7256–7265, 2021.

[7] X. Wu, D. Hong, and J. Chanussot, "UIU-Net: U-Net in U-Net for infrared small object detection," *IEEE Trans. Image Process.*, vol. 32, pp. 364–376, 2023.

[8] M. Mahdianpari et al., "A large-scale change monitoring of wetlands using time series Landsat imagery on Google Earth engine: A case study in Newfoundland," *GISci. Remote Sens.*, vol. 57, no. 8, pp. 1102–1124, Nov. 2020.

[9] Q. Zheng, Q. Weng, and K. Wang, "Characterizing urban land changes of 30 global megacities using nighttime light time series stacks," *ISPRS J. Photogramm. Remote Sens.*, vol. 173, pp. 10–23, Mar. 2021.

[10] J. R. Anderson, *A Land Use and Land Cover Classification System for Use With Remote Sensor Data*, vol. 964. Washington, DC, USA: U.S. Government Printing Office, 1976.

[11] S. Veraverbeke et al., "Hyperspectral remote sensing of fire: State-of-the-art and future perspectives," *Remote Sens. Environ.*, vol. 216, pp. 105–121, Oct. 2018.

[12] D. Tuia, R. Flamary, and N. Courty, "Multiclass feature learning for hyperspectral image classification: Sparse and hierarchical solutions," *ISPRS J. Photogramm. Remote Sens.*, vol. 105, pp. 272–285, Jul. 2015.

[13] P.-H. Hsu, "Feature extraction of hyperspectral images using wavelet and matching pursuit," *ISPRS J. Photogramm. Remote Sens.*, vol. 62, no. 2, pp. 78–92, Jun. 2007.

[14] P. Ghamisi, R. Souza, J. A. Benediktsson, L. Rittner, R. Lotufo, and X. X. Zhu, "Hyperspectral data classification using extended extinction profiles," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 11, pp. 1641–1645, Nov. 2016.

[15] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "CoSpace: Common subspace learning from hyperspectral-multispectral correspondences," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4349–4359, Jul. 2019.

[16] X. Wu, D. Hong, and J. Chanussot, "Convolutional neural networks for multimodal remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5517010.

[17] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615, doi: 10.1109/TGRS.2021.3130716.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:43:18 UTC from IEEE Xplore. Restrictions apply.

[18] Z. Xue, X. Tan, X. Yu, B. Liu, A. Yu, and P. Zhang, "Deep hierarchical vision transformer for hyperspectral and LiDAR data classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3095–3110, 2022.

[19] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "A new deep convolutional neural network for fast hyperspectral image classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 120–147, Nov. 2018.

[20] R. Hang, F. Zhou, Q. Liu, and P. Ghamisi, "Classification of hyperspectral images via multitask generative adversarial networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1424–1436, Feb. 2021.

[21] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.

[22] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 6000–6010.

[23] R. Hang, Z. Li, Q. Liu, P. Ghamisi, and S. S. Bhattacharyya, "Hyperspectral image classification with attention-aided CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2281–2293, Mar. 2021.

[24] J. Lin, L. Zhao, S. Li, R. Ward, and Z. J. Wang, "Active-learning-incorporated deep transfer learning for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 11, pp. 4048–4062, Nov. 2018.

[25] J. Yao et al., "Semi-active convolutional neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5537915, doi: 10.1109/TGRS.2022.3206208.

[26] D. Hong, J. Hu, J. Yao, J. Chanussot, and X. X. Zhu, "Multimodal remote sensing benchmark datasets for land cover classification with a shared and specific feature learning model," *ISPRS J. Photogramm. Remote Sens.*, vol. 178, pp. 68–80, Aug. 2021.

[27] A. Pournemat, P. Adibi, and J. Chanussot, "Semisupervised charting for spectral multimodal manifold learning and alignment," *Pattern Recognit.*, vol. 111, Mar. 2021, Art. no. 107645.

[28] W. Liao, A. Pižurica, R. Bellens, S. Gautama, and W. Philips, "Generalized graph-based fusion of hyperspectral and LiDAR data using morphological features," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 3, pp. 552–556, Mar. 2015.

[29] H. Li, P. Ghamisi, U. Soergel, and X. Zhu, "Hyperspectral and LiDAR fusion using deep three-stream convolutional neural networks," *Remote Sens.*, vol. 10, no. 10, p. 1649, Oct. 2018.

[30] D. Hong et al., "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, May 2021.

[31] D. Hong, J. Yao, D. Meng, Z. Xu, and J. Chanussot, "Multimodal GANs: Toward crossmodal hyperspectral–multispectral image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5103–5113, Jun. 2021.

[32] R. Zhang, P. Isola, and A. A. Efros, "Split-brain autoencoders: Unsupervised learning by cross-channel prediction," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 645–654.

[33] Y. Li, Y. Zhou, Y. Zhang, L. Zhong, J. Wang, and J. Chen, "DKDFN: Domain knowledge-guided deep collaborative fusion network for multimodal unitemporal remote sensing land cover classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 186, pp. 170–189, Apr. 2022.

[34] W. Ma et al., "A collaborative correlation-matching network for multimodality remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5409718.

[35] Z. Xue, X. Yu, A. Yu, B. Liu, P. Zhang, and S. Wu, "Self-supervised feature learning for multimodal remote sensing image land cover classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5533815.

[36] Y. Qing, W. Liu, L. Feng, and W. Gao, "Improved transformer net for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 11, p. 2216, Jun. 2021.

[37] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.

[38] A. Jamali and M. Mahdianpari, "Swin transformer and deep convolutional neural networks for coastal wetland classification using Sentinel-1, Sentinel-2, and LiDAR data," *Remote Sens.*, vol. 14, no. 2, p. 359, Jan. 2022.

[39] Z. Zhao, D. Hu, H. Wang, and X. Yu, "Convolutional transformer network for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[40] H. Dong, L. Zhang, and B. Zou, "Exploring vision transformers for polarimetric SAR image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5219715.

[41] S. K. Roy, A. Deria, D. Hong, B. Rasti, A. Plaza, and J. Chanussot, "Multimodal fusion transformer for remote sensing image classification," 2022, *arXiv:2203.16952*.

[42] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent.*, 2021. [Online]. Available: https://openreview.net/forum?id=YicbFdNTTy

[43] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1800–1807.

[44] D. Hendrycks and K. Gimpel, "Gaussian error linear units (GELUs)," 2016, *arXiv:1606.08415*.

[45] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," *J. Mach. Learn. Res.*, vol. 15, pp. 1929–1958, Jun. 2014.

[46] J. Yao, D. Hong, J. Chanussot, D. Meng, X. Zhu, and Z. Xu, "Cross-attention in coupled unmixing nets for unsupervised hyperspectral super-resolution," in *Proc. 16th Eur. Conf. Comput. Vis.*, pp. 208–224, Springer, 2020.

[47] H. Zhang, J. Yao, L. Ni, L. Gao, and M. Huang, "Multimodal attention-aware convolutional neural networks for classification of hyperspectral and LiDAR data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 3635–3644, 2023.

[48] D. Hong, L. Gao, R. Hang, B. Zhang, and J. Chanussot, "Deep encoder–decoder networks for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[49] D. Hong, N. Yokoya, J. Chanussot, J. Xu, and X. X. Zhu, "Learning to propagate labels on graphs: An iterative multitask regression framework for semi-supervised hyperspectral dimensionality reduction," *ISPRS J. Photogramm. Remote Sens.*, vol. 158, pp. 35–49, Dec. 2019.

[50] A. Okujeni, S. van der Linden, and P. Hostert, "Berlin-urban-gradient dataset 2009—An EnMAP preparatory flight campaign (datasets)," GFZ Data Services, Germany, 2016, pp. 1–9, doi: 10.5880/enmap.2016.002.

[51] M. Haklay and P. Weber, "OpenStreetMap: User-generated street maps," *IEEE Pervasive Comput.*, vol. 7, no. 4, pp. 12–18, Oct. 2008.

[52] J. Hu et al., "MDAS: A new multimodal benchmark dataset for remote sensing," *Earth Syst. Sci. Data*, vol. 15, no. 1, pp. 113–131, Jan. 2023.

[53] K. Segl et al., "Simulation of spatial sensor characteristics in the context of the EnMAP hyperspectral mission," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 7, pp. 3046–3054, Jul. 2010.

[54] J. Cohen, "A coefficient of agreement for nominal scales," *Educ. Psychol. Meas.*, vol. 20, no. 1, pp. 37–46, Apr. 1960.

[55] M. Belgiu and L. Drăguţ, "Random forest in remote sensing: A review of applications and future directions," *ISPRS J. Photogramm. Remote Sens.*, vol. 114, pp. 24–31, Apr. 2016.

[56] D. Hong, N. Yokoya, N. Ge, J. Chanussot, and X. X. Zhu, "Learnable manifold alignment (LeMA): A semi-supervised cross-modality learning framework for land cover and land use classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 147, pp. 193–205, Jan. 2019.

[57] R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia, and Q. Liu, "Classification of hyperspectral and LiDAR data using coupled CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4939–4950, Jul. 2020.

Jing Yao (Member, IEEE) received the Ph.D. degree in mathematics from Xi'an Jiaotong University, Xi'an, China, in 2021.

From 2019 to 2020, he was a Visiting Student at the Signal Processing in Earth Observation (SiPEO), Technical University of Munich (TUM), Munich, Germany, and the Remote Sensing Technology Institute (IMF), German Aerospace Center (DLR), Oberpfaffenhofen, Germany. He is currently an Assistant Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China. His research interests include hyperspectral and multimodal remote sensing image analysis, mainly including optimization and deep-learning-based methods for image processing and interpretation tasks.

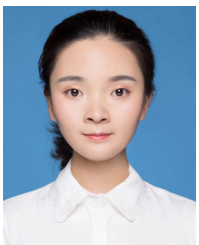
Dr. Yao was a recipient of the Jose Bioucas Dias Award for recognizing the outstanding paper at the Workshop on Hyperspectral Imaging and Signal Processing: Evolution in Remote Sensing (WHISPERS) in 2021. He serves as a Guest Editor for *Remote Sensing*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:43:18 UTC from IEEE Xplore. Restrictions apply.

Bing Zhang (Fellow, IEEE) received the B.S. degree in geography from Peking University, Beijing, China, in 1991, and the M.S. and Ph.D. degrees in remote sensing from the Institute of Remote Sensing Applications, Chinese Academy of Sciences (CAS), Beijing, in 1994 and 2003, respectively.

He is currently a Full Professor and the Deputy Director of the Aerospace Information Research Institute, CAS, where he has been leading lots of key scientific projects in the area of hyperspectral remote sensing for more than 25 years. He has developed five software systems in image processing and applications. He has authored more than 300 publications, including more than 170 journal articles. He has edited six books/contributed book chapters on hyperspectral image processing and subsequent applications. His research interests include the development of mathematical and physical models and image processing software for the analysis of hyperspectral remote sensing data in many different areas.

Dr. Zhang has been serving as a Technical Committee Member for the IEEE Workshop on Hyperspectral Image and Signal Processing since 2011, the President for the Hyperspectral Remote Sensing Committee of the China National Committee of the International Society for Digital Earth since 2012, and the Standing Director for the Chinese Society of Space Research (CSSR) since 2016. He is the Student Paper Competition Committee Member of IGARSS from 2015 to 2019. He has received the National Science Foundation for Distinguished Young Scholars of China in 2013 and the 2016 Outstanding Science and Technology Achievement Prize of the Chinese Academy of Sciences, the highest level of awards for CAS scholars. His creative achievements were rewarded with ten important prizes from the Chinese government and special government allowances from the Chinese State Council. He has been serving as an Associate Editor for the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS).



Chenyu Li (Student Member, IEEE) received the B.S. and M.S. degrees from the School of Transportation, Southeast University (SEU), Nanjing, China, in 2018 and 2021, respectively. She is currently pursuing the Ph.D. degree with the Department of Mathematics, SEU. She is also a co-supervised Ph.D. Student at the Aerospace Information Research Institute, Chinese Academy of Sciences (CAS), Beijing, China.

Her research interests include hyperspectral remote sensing, interpretable artificial intelligence, and target detection.

Danfeng Hong (Senior Member, IEEE) received the M.Sc. degree (summa cum laude) in computer vision from the College of Information Engineering, Qingdao University, Qingdao, China, in 2015, and the Dr.-Ing. degree (summa cum laude) from the Signal Processing in Earth Observation (SiPEO), Technical University of Munich (TUM), Munich, Germany, in 2019.

He is currently a Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences (CAS), Beijing, China. Before joining CAS, he was the Research Scientist and led the Spectral Vision Working Group, Remote Sensing Technology Institute (IMF), German Aerospace Center (DLR), Oberpfaffenhofen, Germany. He was also the Adjunct Scientist of the GIPSA-Laboratory, Grenoble INP, CNRS, Université Grenoble Alpes, Grenoble, France. His research interests include artificial intelligence, remote sensing big data analysis, multimodal interpretation, and their applications in Earth vision.

Dr. Hong was a recipient of the Best Reviewer Award of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS) in 2021 and 2022, the Jose Bioucas Dias Award for recognizing the outstanding paper at Workshop on Hyperspectral Imaging and Signal Processing: Evolution in Remote Sensing (WHISPERS) in 2021, the Best Reviewer Award of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (JSTARS) in 2022, the Remote Sensing Young Investigator Award in 2022, the IEEE GRSS Early Career Award in 2022, and a Highly Cited Researcher (Clarivate Analytics/Thomson Reuters) in 2022. He is an Associate Editor of the IEEE TGRS, an Editorial Board Member of *Remote Sensing*, and an Editorial Advisory Board Member of the *ISPRS Journal of Photogrammetry and Remote Sensing*.

Jocelyn Chanussot (Fellow, IEEE) received the M.Sc. degree in electrical engineering from the Grenoble Institute of Technology (Grenoble INP), Grenoble, France, in 1995, and the Ph.D. degree from the Université de Savoie, Annecy, France, in 1998.

From 2015 to 2017, he was a Visiting Professor with the University of California at Los Angeles (UCLA), Los Angeles, CA, USA. Since 1999, he has been with Grenoble INP, where he is currently a Professor of signal and image processing. He has been a Visiting Scholar with Stanford University, Stanford, CA, USA; the KTH Royal Institute of Technology (KTH), Stockholm, Sweden; and the National University of Singapore (NUS), Singapore. Since 2013, he has been an Adjunct Professor with the University of Iceland, Reykjavik, Iceland. He holds the AXA Chair in remote sensing and is an Adjunct Professor at the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China. His research interests include image analysis, hyperspectral remote sensing, data fusion, machine learning, and artificial intelligence.

Dr. Chanussot was a member of the Machine Learning for Signal Processing Technical Committee of the IEEE Signal Processing Society from 2006 to 2008. He is a member of the Institut Universitaire de France from 2012 to 2017. He was the Founding President of the IEEE Geoscience and Remote Sensing French Chapter from 2007 to 2010, which has received the 2010 IEEE GRS-S Chapter Excellence Award. He has received multiple outstanding paper awards. He was the Vice-President of the IEEE Geoscience and Remote Sensing Society, in charge of meetings and symposia, from 2017 to 2019. He was the General Chair of the first IEEE GRSS Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing (WHISPERS). He was the Co-Chair of the GRS Data Fusion Technical Committee from 2005 to 2008 and the Chair from 2009 to 2011. He was the Program Chair of the IEEE International Workshop on Machine Learning for Signal Processing in 2009. He was the Editor-in-Chief of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING from 2011 to 2015. In 2014, he has served as a Guest Editor for *IEEE Signal Processing Magazine*. He is an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTIONS ON IMAGE PROCESSING, and the PROCEEDINGS OF THE IEEE. He is also a Highly Cited Researcher (Clarivate Analytics/Thomson Reuters).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:43:18 UTC from IEEE Xplore. Restrictions apply.